SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

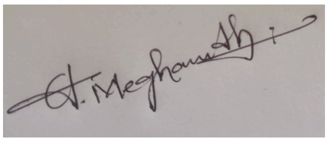
QUESTIONS: **Total Marks: 20**

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I _A.Meghanadh(192372281)___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. **SOAP (Simple Object Access Protocol):**
   SOAP is a protocol used for exchanging structured information between applications over a network, usually using XML.

2. **WSDL (Web Services Description Language):**
   WSDL is an XML-based language that describes a web service, including its operations, messages, data types, and endpoint.

3. **API (Application Programming Interface):**
   An API is a set of rules and methods that allows one software application to communicate with and use the functions of another application.

4. **XML Schema:**
   XML Schema (XSD) defines the structure, elements, attributes, and data types allowed in an XML document.

5. **Web Service:**
   A web service is a software system that allows different applications to communicate and exchange data over a network using standard web technologies.

6. **REST (Representational State Transfer):**
   REST is an architectural style for designing web services that uses HTTP methods such as GET, POST, PUT, and DELETE to work with resources.

7. **Protocol:**
   A protocol is a set of rules that defines how data is transmitted and communicated between devices or applications.

8. **Endpoint in Web Services:**
   An endpoint is a specific URL or network address through which a web service can be accessed.

9. **Service in Web Applications:**
   A service is a software component that provides specific functionality or data to other applications or components.

10. **Difference between SOAP and REST:**

SOAP	REST
Protocol	Architectural style
Primarily uses XML	Commonly uses JSON, but can use XML and others
More structured and strict	Simpler and more flexible
Uses SOAP messages	Uses standard HTTP methods
Generally more overhead	Generally lightweight

11. **Role of WSDL in Web Services:**
   WSDL describes how a web service works and tells clients what operations are available, what data is required, and where the service can be accessed.

12. **JSON and its use in APIs:**
   JSON (JavaScript Object Notation) is a lightweight data format used to represent and exchange data. REST APIs commonly use JSON for sending requests and responses.

13. **HTTP and its use in Web Services:**
   HTTP (Hypertext Transfer Protocol) is a communication protocol used to transfer data over the web. Web services use HTTP to send requests and receive responses between clients and servers.

14. **Request and Response in Web Communication:**
   A request is a message sent by a client to a server asking for data or an

operation. A response is the server's message containing the requested data, result, or an error.

15. **Importance of XML in Web Services:**
    XML provides a standardized, platform-independent way to structure and exchange data. It is widely used in SOAP messages and WSDL documents.

16. **Purpose of an API Endpoint:**
    An API endpoint provides a specific URL through which a client can access a particular resource or operation of an API.

17. **Difference between HTTP and HTTPS:**
    HTTP transfers data without encryption. HTTPS uses encryption through TLS/SSL to protect data during communication and provides better security.

18. **RESTful Services:**
    RESTful services are web services designed according to REST principles. They use resources identified by URLs and standard HTTP methods such as GET, POST, PUT, and DELETE.

19. **Namespace in XML:**
    An XML namespace is used to uniquely identify XML elements and attributes and prevent naming conflicts between different XML vocabularies.

20. **Role of Client and Server in Web Services:**
    The client sends requests to access a service or data. The server receives the request, processes it, and sends an appropriate response back to the client.